Experiment Report - classification

19 configurations of 4 model(s), ranked by accuracy (higher is better) · 2026-11-11 22:39

1. Executive Summary

best configuration	LogisticRegression (rank #1)
accuracy	0.9667 (higher is better)
all metrics	accuracy 0.9667, precision 0.9678, recall 0.9667, f1 0.9666
hyper-parameters	{"C": 10}
features	4: sepal length (cm), sepal width (cm), petal length (cm), petal width (cm)
validation	k_folds
configurations / models	19 / 4
visualizations	23 generated, 3 skipped, 0 failed

- LogisticRegression leads SVC by 0.0000 accuracy (best configuration of each).
- LinearSVC varies most across its configurations (std of accuracy 0.0482).
- Always predicting the most frequent class would score accuracy 0.3333; the best model scores 0.9667.
- Training time per configuration: median 0.031 s, total 2.2 s of worker time.
- The best of many configurations is optimistically biased; confirm on held-out data.

2. Experiment Configuration

created	2026-11-11 22:39:15
task	classification
validation	k_folds
ranking metric	accuracy (higher is better)
models	LogisticRegression, SVC, RandomForestClassifier, LinearSVC
preprocessing	StandardScaler()
feature subsets tried	1
grid: max_depth	[2, 4, null]
grid: C	[0.01, 0.1, 1, 10]
grid: kernel	["rbf", "linear"]

3. Dataset Overview

rows	150

features	4

Feature columns: sepal length (cm), sepal width (cm), petal length (cm), petal width (cm)

4. Overall Results and Model Comparison

Per model

Best, mean, spread and worst accuracy over each model's configurations, and the mean of every other metric.

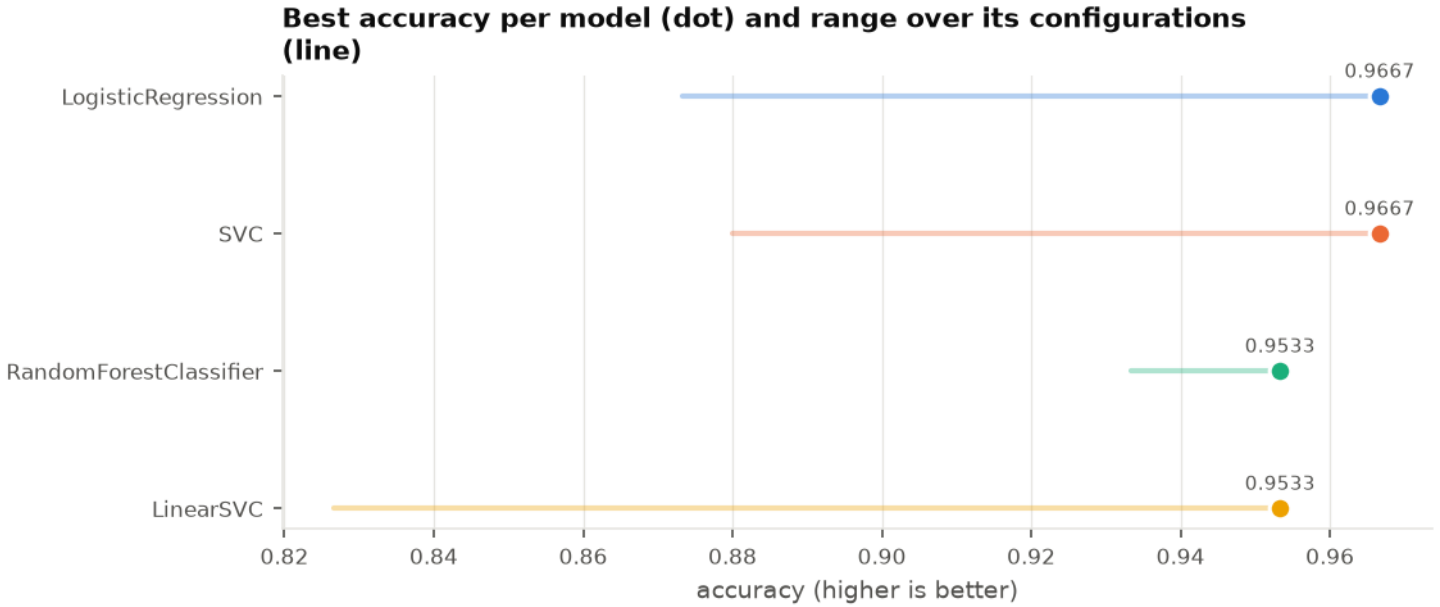
model	configurations	best_accuracy	mean_accuracy	std_accuracy	worst_accuracy	mean_precision	mean_recall	mean_f1	best_rank	best_n_features
LogisticRegression	4	0.9667	0.9300	0.0373	0.8733	0.9333	0.9300	0.9297	1	4
SVC	8	0.9667	0.9358	0.0309	0.8800	0.9366	0.9358	0.9358	2	4
RandomForestClassifier	3	0.9533	0.9444	0.0083	0.9333	0.9446	0.9444	0.9444	6	4
LinearSVC	4	0.9533	0.9050	0.0482	0.8267	0.9109	0.9050	0.9040	9	4

Top 19 configurations

rank	model	params	n_features	validation	accuracy	precision	recall	f1
1	LogisticRegression	{"C": 10}	4	k_folds	0.9667	0.9678	0.9667	0.9666
2	SVC	{"C": 1, "kernel": "linear"}	4	k_folds	0.9667	0.9678	0.9667	0.9666
3	LogisticRegression	{"C": 1}	4	k_folds	0.9600	0.9605	0.9600	0.9600
4	SVC	{"C": 0.1, "kernel": "linear"}	4	k_folds	0.9600	0.9619	0.9600	0.9599
5	SVC	{"C": 10, "kernel": "linear"}	4	k_folds	0.9600	0.9605	0.9600	0.9600
6	RandomForestClassifier	{"max_depth": 4}	4	k_folds	0.9533	0.9534	0.9533	0.9533
7	SVC	{"C": 1, "kernel": "rbf"}	4	k_folds	0.9533	0.9539	0.9533	0.9535
8	SVC	{"C": 10, "kernel": "rbf"}	4	k_folds	0.9533	0.9544	0.9533	0.9533
9	LinearSVC	{"C": 10}	4	k_folds	0.9533	0.9539	0.9533	0.9535
10	RandomForestClassifier	{"max_depth": null}	4	k_folds	0.9467	0.9467	0.9467	0.9467
11	RandomForestClassifier	{"max_depth": 2}	4	k_folds	0.9333	0.9338	0.9333	0.9333
12	LinearSVC	{"C": 1}	4	k_folds	0.9333	0.9340	0.9333	0.9336
13	LogisticRegression	{"C": 0.1}	4	k_folds	0.9200	0.9213	0.9200	0.9202
14	SVC	{"C": 0.1, "kernel": "rbf"}	4	k_folds	0.9133	0.9142	0.9133	0.9133
15	LinearSVC	{"C": 0.1}	4	k_folds	0.9067	0.9092	0.9067	0.9068
16	SVC	{"C": 0.01, "kernel": "rbf"}	4	k_folds	0.9000	0.9001	0.9000	0.9000
17	SVC	{"C": 0.01, "kernel": "linear"}	4	k_folds	0.8800	0.8803	0.8800	0.8800
18	LogisticRegression	{"C": 0.01}	4	k_folds	0.8733	0.8838	0.8733	0.8718
19	LinearSVC	{"C": 0.01}	4	k_folds	0.8267	0.8465	0.8267	0.8222

Model ranking

id: model_ranking · model(s): LogisticRegression (rank #1), SVC (rank #2), RandomForestClassifier (rank #6), LinearSVC (rank #9)

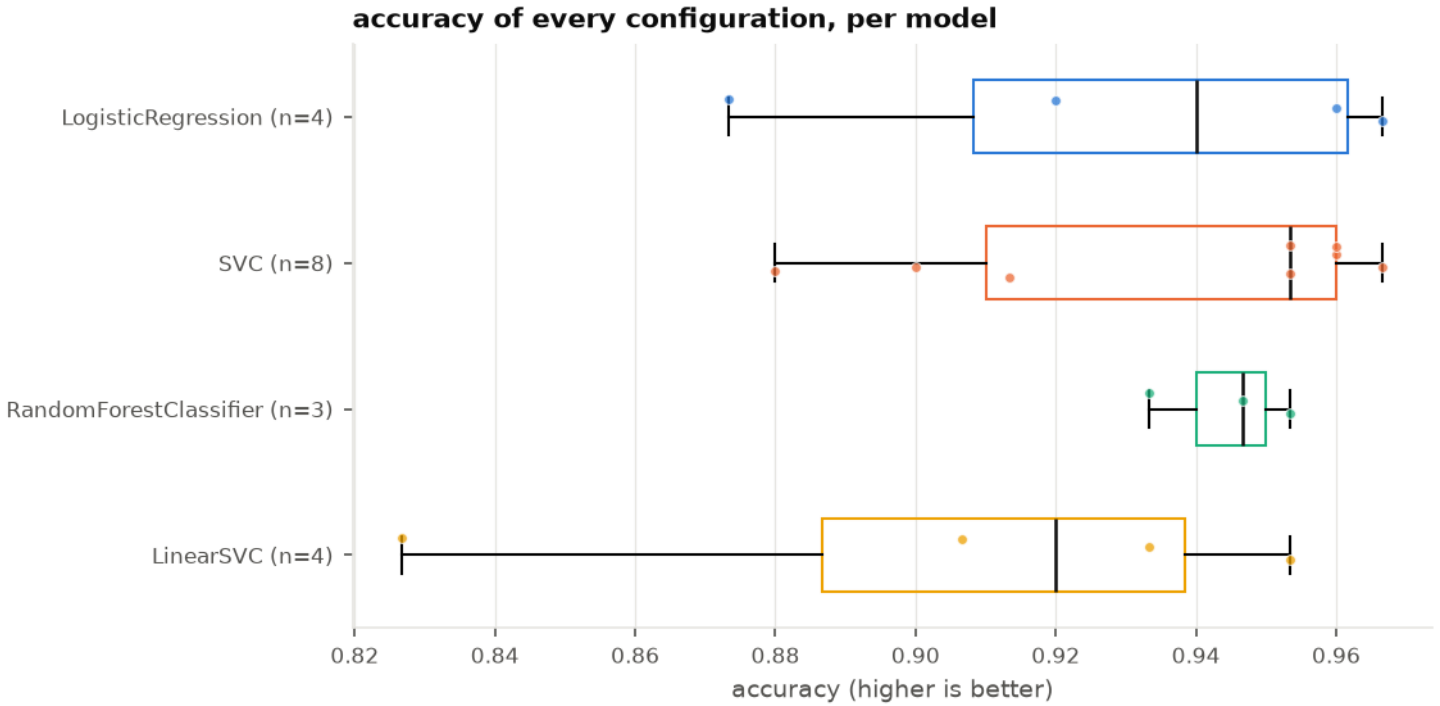


What it shows. Each model's best configuration (dot) and the range of the ranking metric over all of its configurations (line), best model on top. A long line means the model is sensitive to its hyper-parameters / feature subset. The best of many tries is optimistically biased: compare with the spread, not only the dot.

LogisticRegression best accuracy	0.9667
LogisticRegression configurations	4
SVC best accuracy	0.9667
SVC configurations	8
RandomForestClassifier best accuracy	0.9533
RandomForestClassifier configurations	3
LinearSVC best accuracy	0.9533
LinearSVC configurations	4

Score distribution across configurations

id: config_distribution · model(s): LogisticRegression (rank #1), SVC (rank #2), RandomForestClassifier (rank #6), LinearSVC (rank #9)

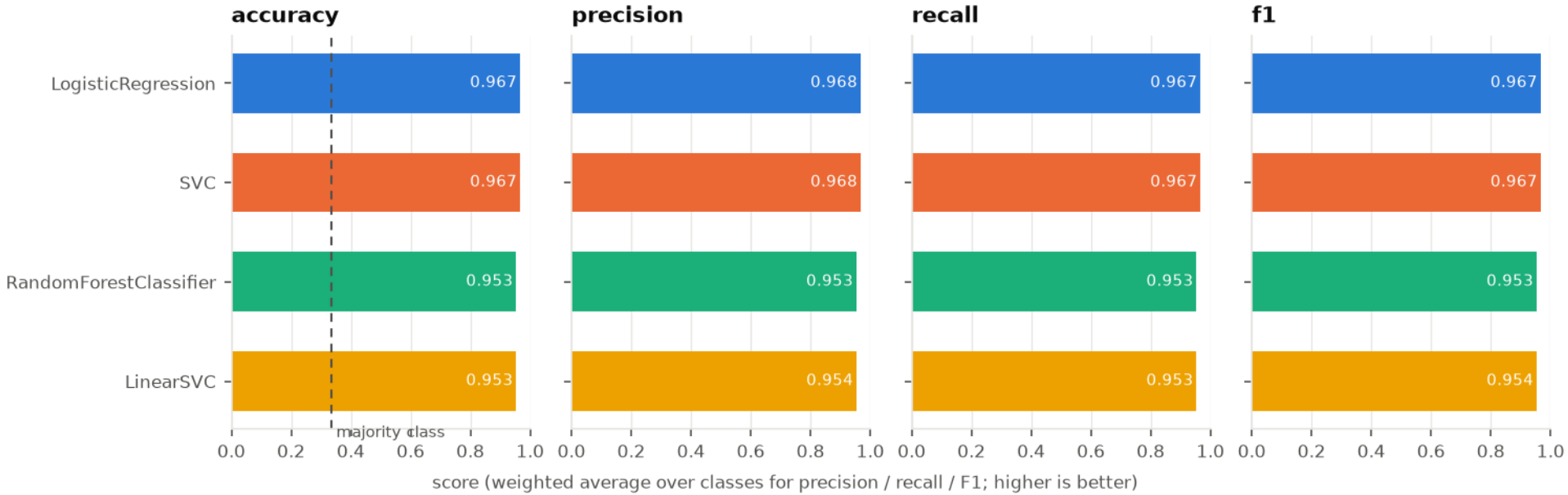


What it shows. Box plot of the ranking metric over every configuration (hyper-parameters x feature subset) of each model, with every configuration drawn as a point. Shows how robust a model family is to its settings - a narrow box near the top is a safer choice than a single lucky configuration.

LogisticRegression median accuracy	0.9400
SVC median accuracy	0.9533
RandomForestClassifier median accuracy	0.9467
LinearSVC median accuracy	0.9200

Classification metrics by model

id: clf_metrics · model(s): LogisticRegression (rank #1), SVC (rank #2), RandomForestClassifier (rank #6), LinearSVC (rank #9)



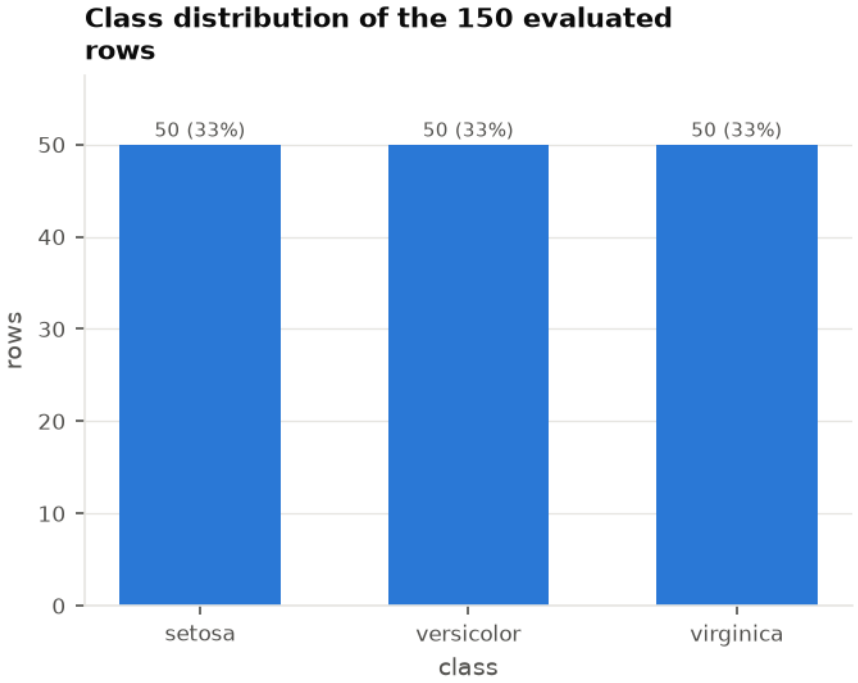
What it shows. Accuracy, precision, recall and F1 of each model's best configuration, one panel per metric (all on the same 0-1 scale). The dashed line is the accuracy of always predicting the most frequent class: a model near it has learned little.

LogisticRegression	accuracy 0.9667, precision 0.9678, recall 0.9667, f1 0.9666
SVC	accuracy 0.9667, precision 0.9678, recall 0.9667, f1 0.9666
RandomForestClassifier	accuracy 0.9533, precision 0.9534, recall 0.9533, f1 0.9533
LinearSVC	accuracy 0.9533, precision 0.9539, recall 0.9533, f1 0.9535
majority-class accuracy (baseline)	0.3333

5. Classification Analysis

Class distribution

id: clf_class_distribution

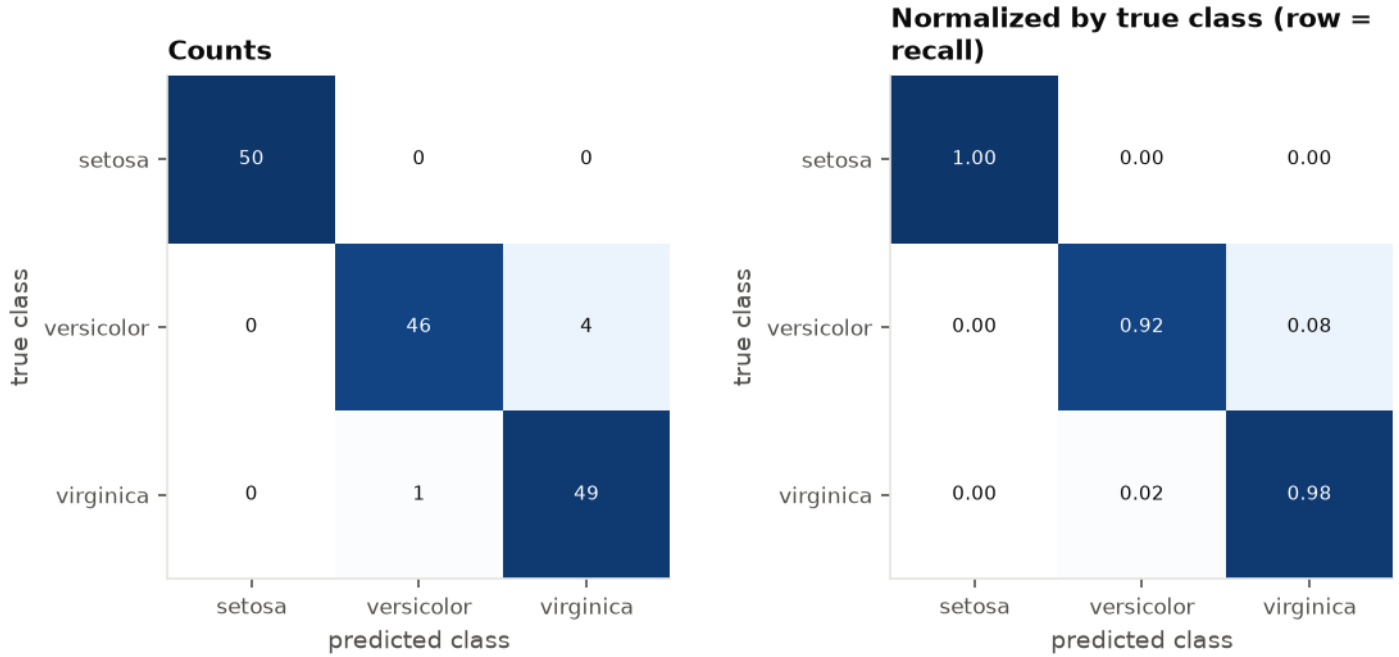


What it shows. Number of rows per class among the rows the scores were computed on (all rows for k-fold / leave-one-out, the test rows for a train / test split). Strong imbalance makes accuracy misleading - read the per-class and precision-recall plots.

classes	3
rows	150
largest / smallest class	1.00

Confusion matrix - LogisticRegression

id: clf_confusion:LogisticRegression · model(s): LogisticRegression (rank #1)

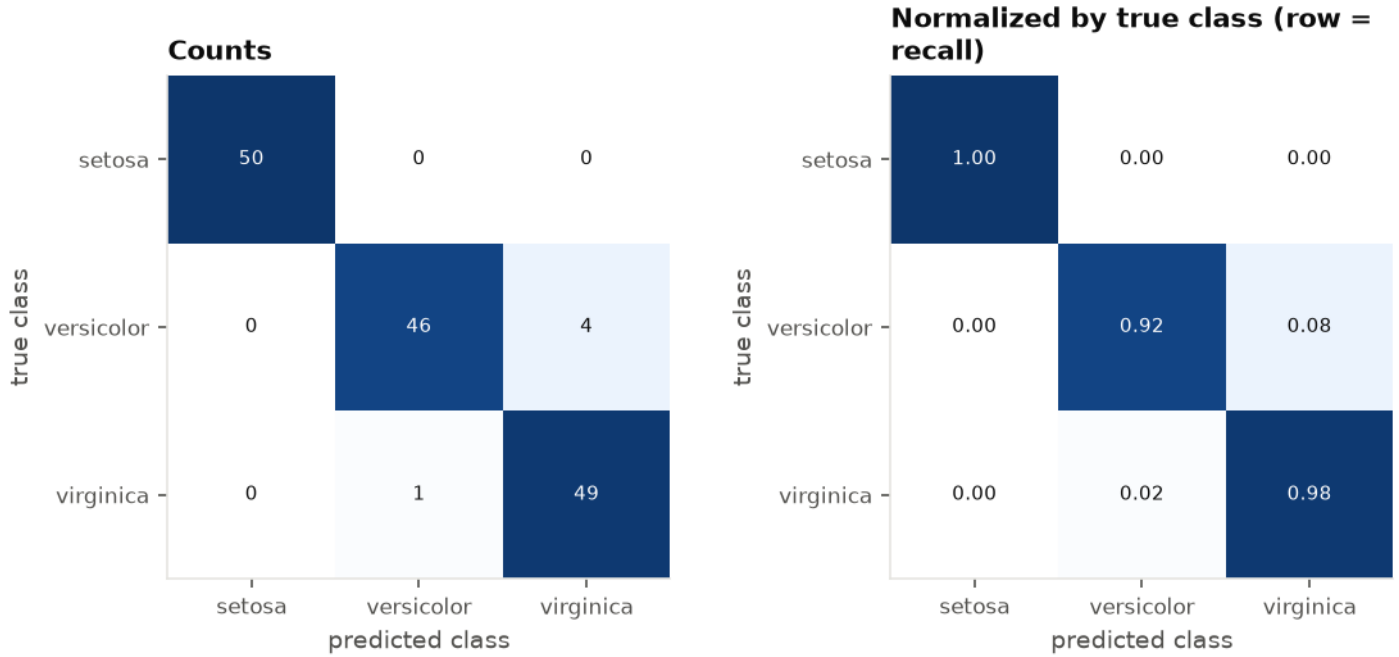


What it shows. Rows are the true classes, columns the predictions. Left: counts. Right: each row divided by its total, so the diagonal is per-class recall and off-diagonal cells show which classes are mistaken for which - independent of class size.

accuracy	0.9667
lowest recall	versicolor (0.920)
rows	150

Confusion matrix - SVC

id: clf_confusion:SVC · model(s): SVC (rank #2)

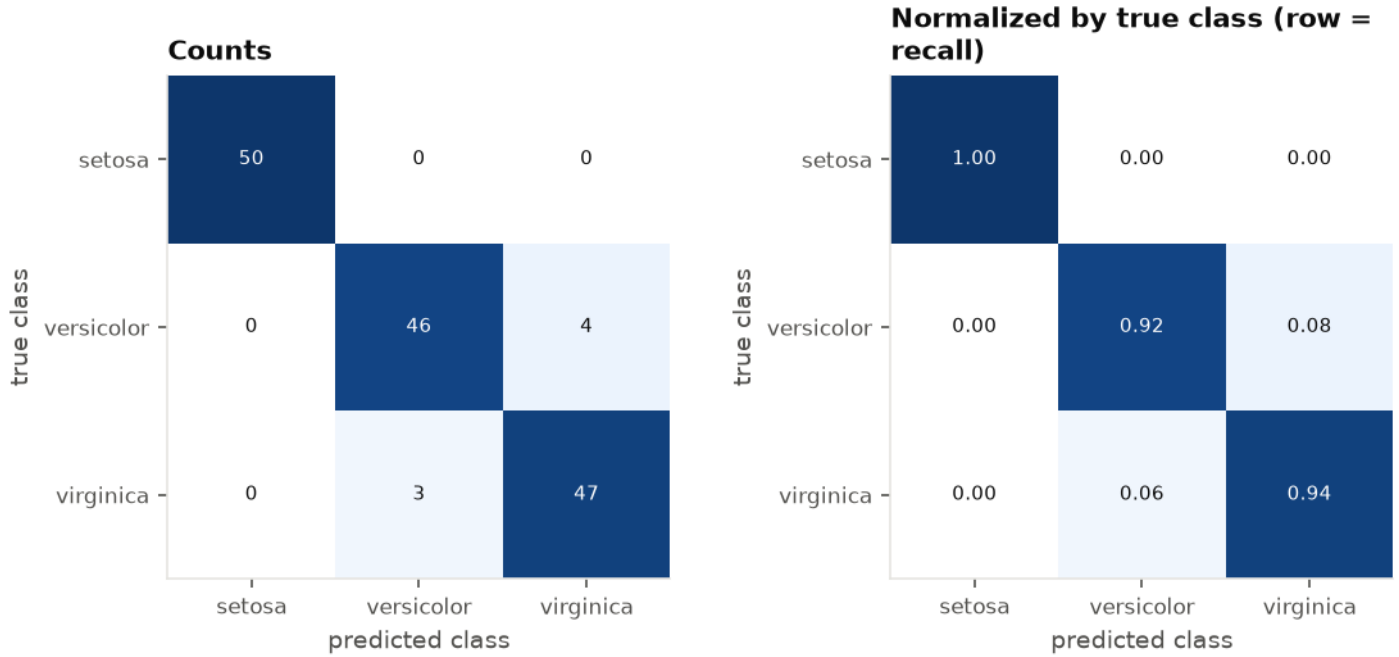


What it shows. Rows are the true classes, columns the predictions. Left: counts. Right: each row divided by its total, so the diagonal is per-class recall and off-diagonal cells show which classes are mistaken for which - independent of class size.

accuracy	0.9667
lowest recall	versicolor (0.920)
rows	150

Confusion matrix - RandomForestClassifier

id: clf_confusion:RandomForestClassifier · model(s): RandomForestClassifier (rank #6)

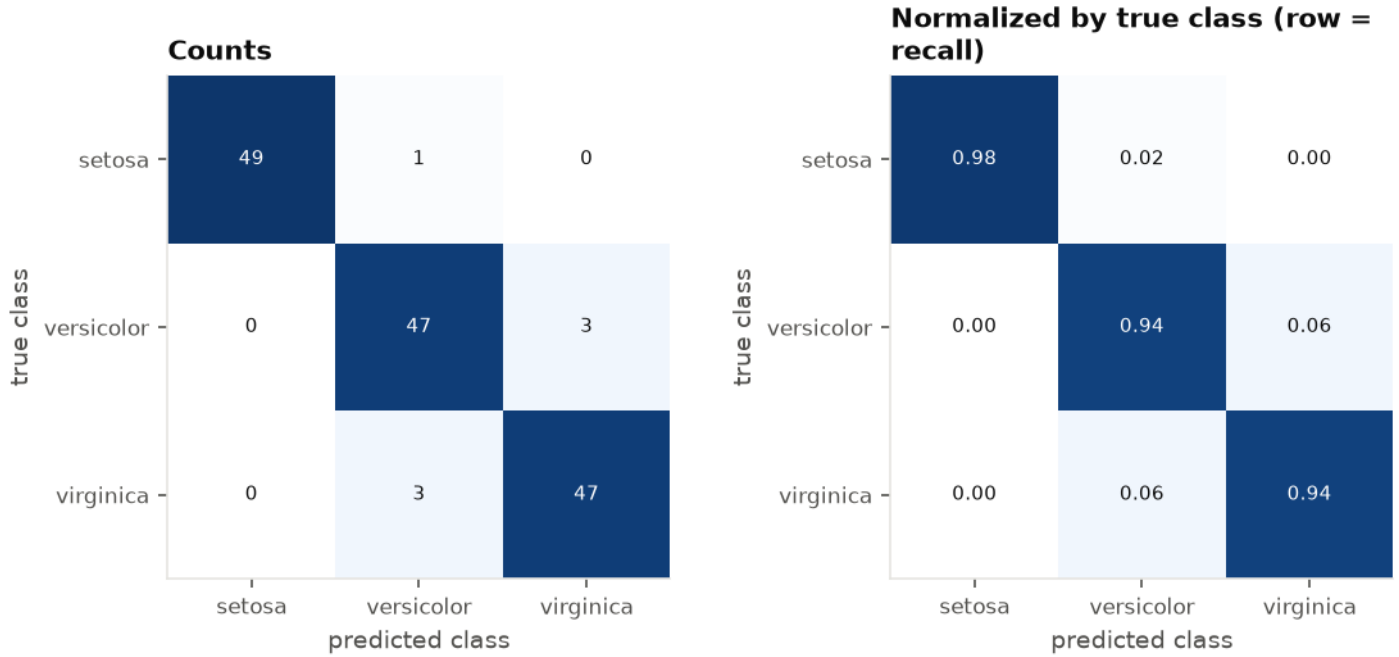


What it shows. Rows are the true classes, columns the predictions. Left: counts. Right: each row divided by its total, so the diagonal is per-class recall and off-diagonal cells show which classes are mistaken for which - independent of class size.

accuracy	0.9533
lowest recall	versicolor (0.920)
rows	150

Confusion matrix - LinearSVC

id: clf_confusion:LinearSVC · model(s): LinearSVC (rank #9)

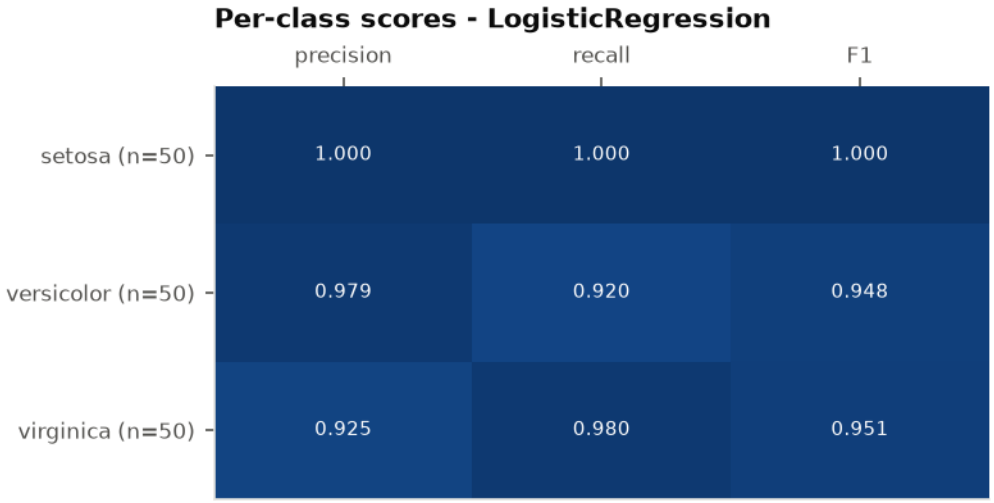


What it shows. Rows are the true classes, columns the predictions. Left: counts. Right: each row divided by its total, so the diagonal is per-class recall and off-diagonal cells show which classes are mistaken for which - independent of class size.

accuracy	0.9533
lowest recall	versicolor (0.940)
rows	150

Per-class precision, recall and F1

id: clf_per_class · model(s): LogisticRegression (rank #1)

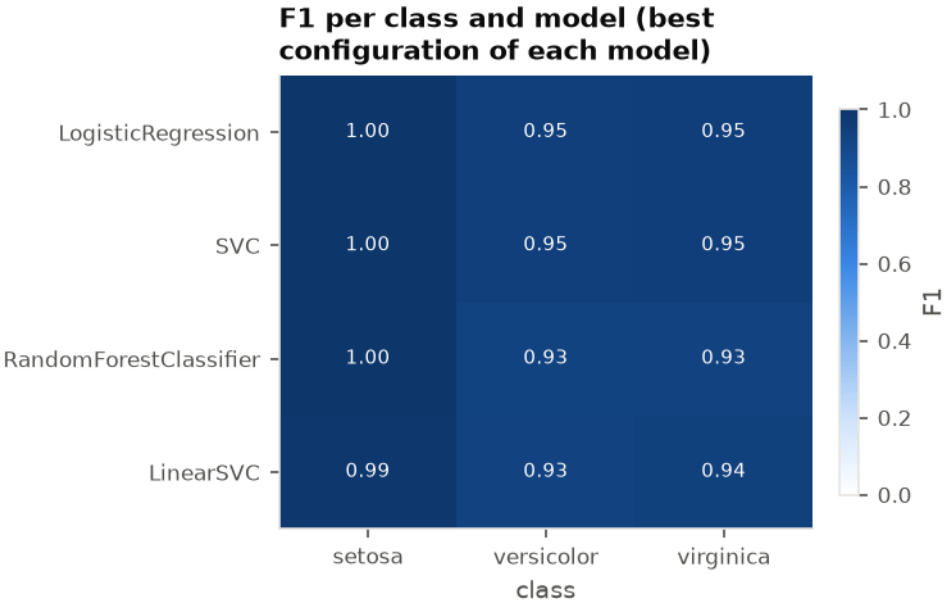


What it shows. Precision, recall and F1 of every class for the best model, with each class's support (n). The weighted averages in the leaderboard can hide a class the model almost never gets right; scores of small classes are noisy.

macro F1	0.9666
weighted F1	0.9666
worst class (F1)	versicolor (0.948)

Per-class F1 across models

id: clf_per_class_models · model(s): LogisticRegression (rank #1), SVC (rank #2), RandomForestClassifier (rank #6), LinearSVC (rank #9)

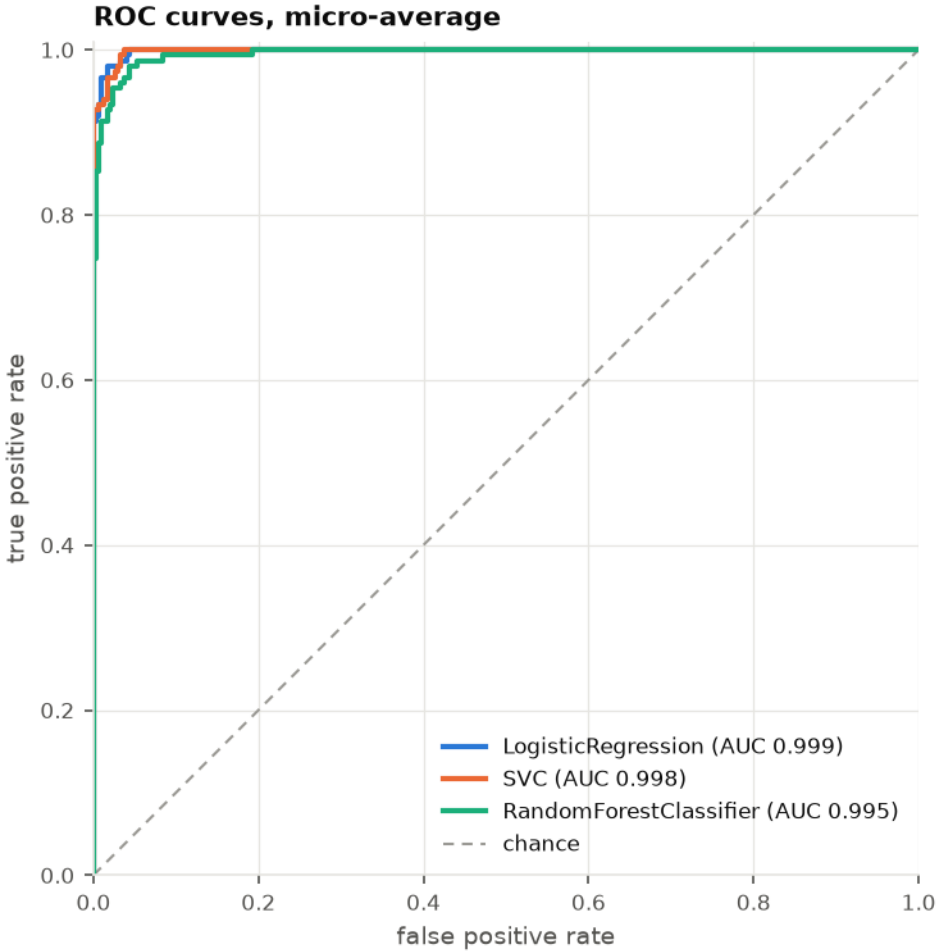


What it shows. F1 of every class for each model. Shows whether models fail on the same classes (a data problem) or on different ones (an ensemble may help).

best on setosa	LogisticRegression
best on versicolor	LogisticRegression
best on virginica	LogisticRegression

ROC curves

id: clf_roc · model(s): LogisticRegression (rank #1), SVC (rank #2), RandomForestClassifier (rank #6)



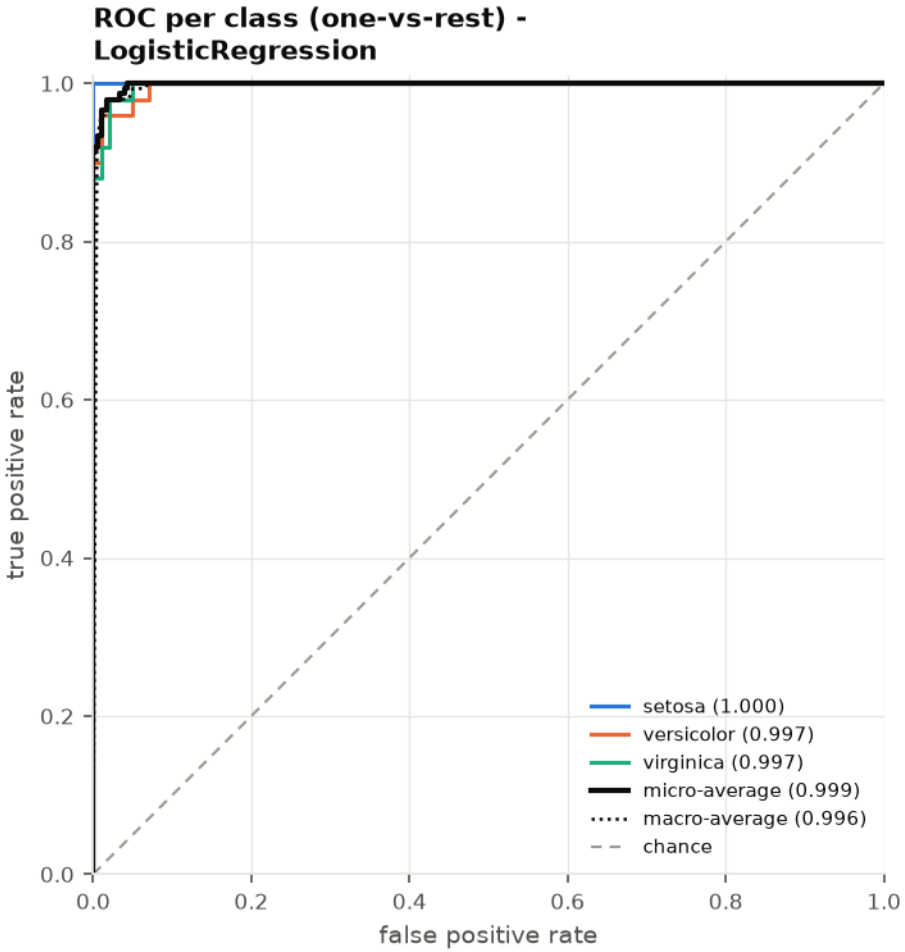
What it shows. True vs false positive rate over every decision threshold, for each model's best configuration (multiclass: micro-average over one-vs-rest classes). AUC is the probability that a random positive is ranked above a random negative; 0.5 is chance. Can look optimistic when positives are rare - see the precision-recall curves.

No predict_proba (not shown): LinearSVC

LogisticRegression ROC AUC	0.9986
SVC ROC AUC	0.9984
RandomForestClassifier ROC AUC	0.9951

ROC per class

id: clf_roc_per_class · model(s): LogisticRegression (rank #1)

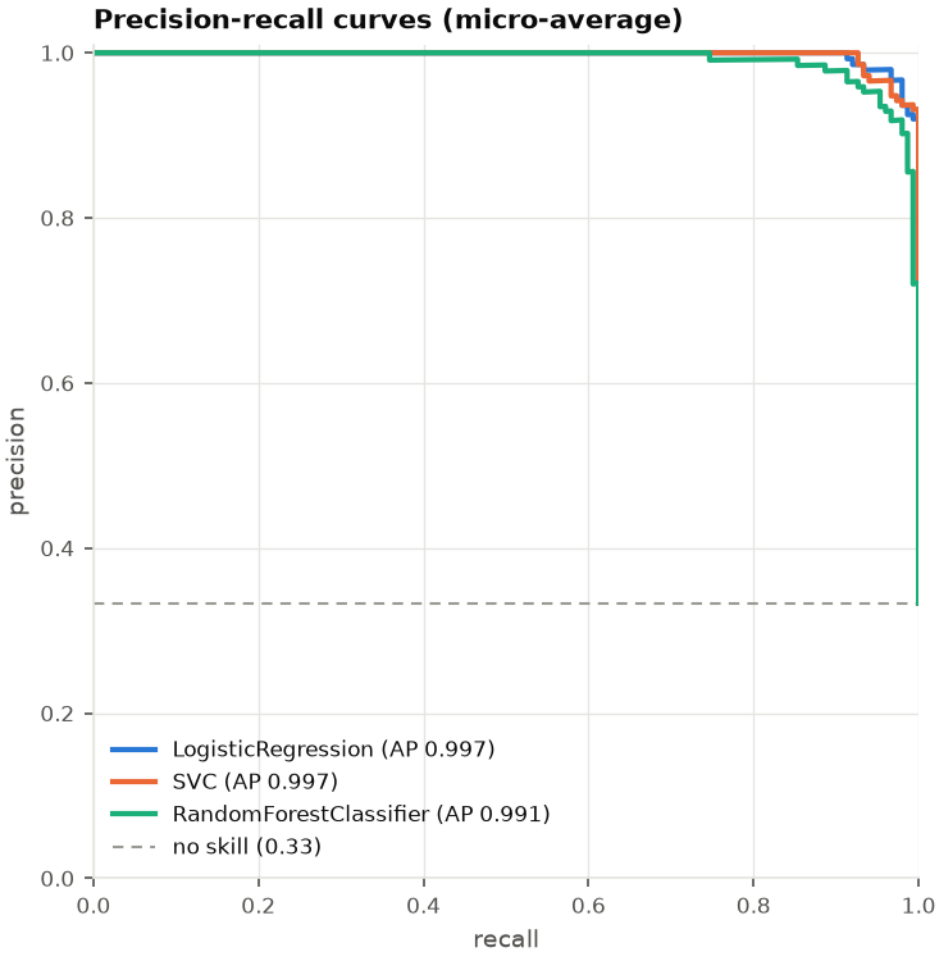


What it shows. One-vs-rest ROC curve for each class of the best model, with the micro-average (every prediction pooled) and the macro-average (classes weighted equally). A class far below the others is the hard one to separate.

class setosa AUC	1.0000
class versicolor AUC	0.9970
class virginica AUC	0.9974
micro-average AUC	0.9986
macro-average AUC	0.9960

Precision-recall curves

id: clf_pr · model(s): LogisticRegression (rank #1), SVC (rank #2), RandomForestClassifier (rank #6)



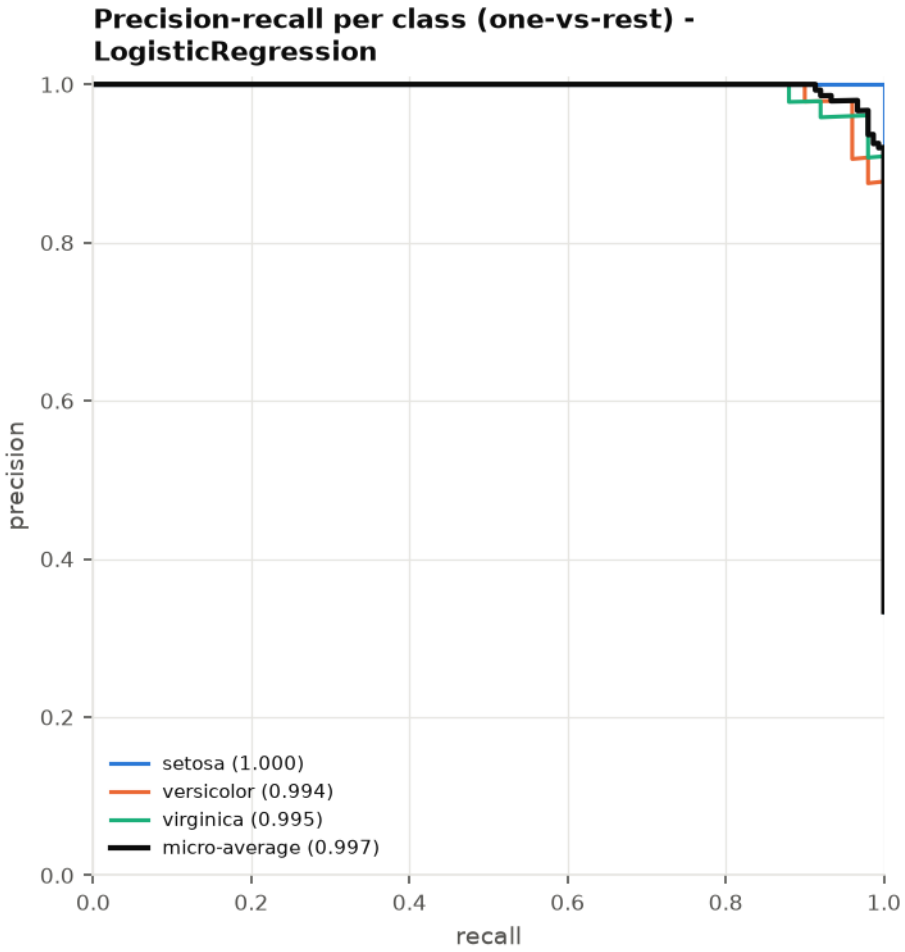
What it shows. Precision against recall over every threshold (multiclass: micro-average). Average precision (AP) summarises the curve; the dashed line is a classifier with no skill (the positive rate). More informative than ROC when the positive class is rare.

No predict_proba (not shown): LinearSVC

LogisticRegression average precision	0.9972
SVC average precision	0.9968
RandomForestClassifier average precision	0.9909
no-skill precision (prevalence)	0.3333

Precision-recall per class

id: clf_pr_per_class · model(s): LogisticRegression (rank #1)



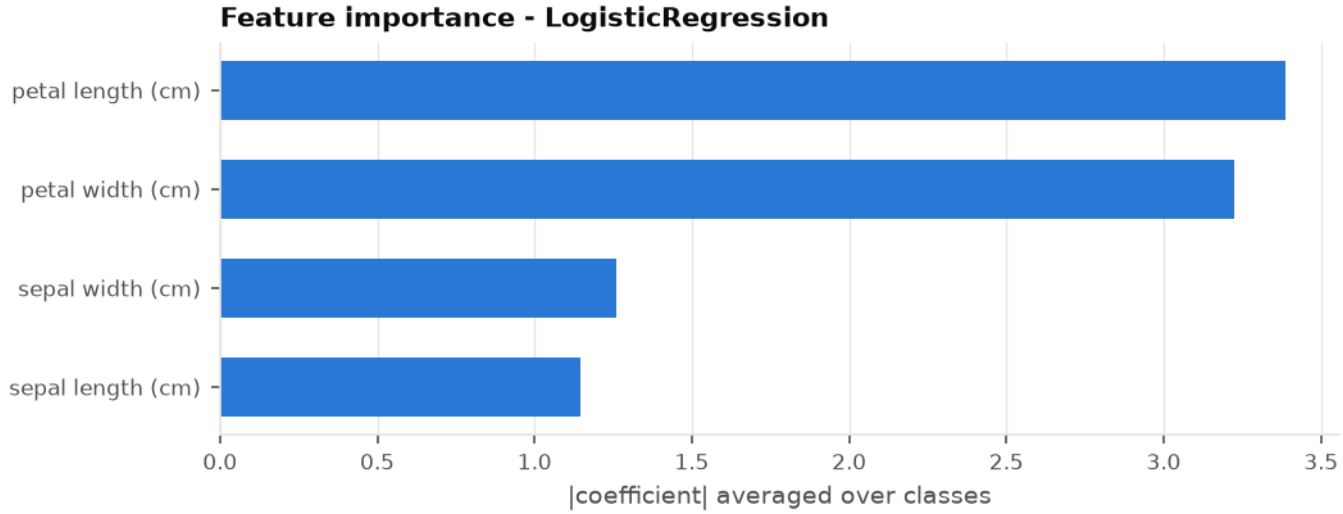
What it shows. One-vs-rest precision-recall curve for each class of the best model, with average precision per class and the micro-average.

class setosa AP	1.0000
class versicolor AP	0.9944
class virginica AP	0.9949
micro-average AP	0.9972

6. Feature Analysis

Feature importance - LogisticRegression

id: feature_importance:LogisticRegression · model(s): LogisticRegression (rank #1)

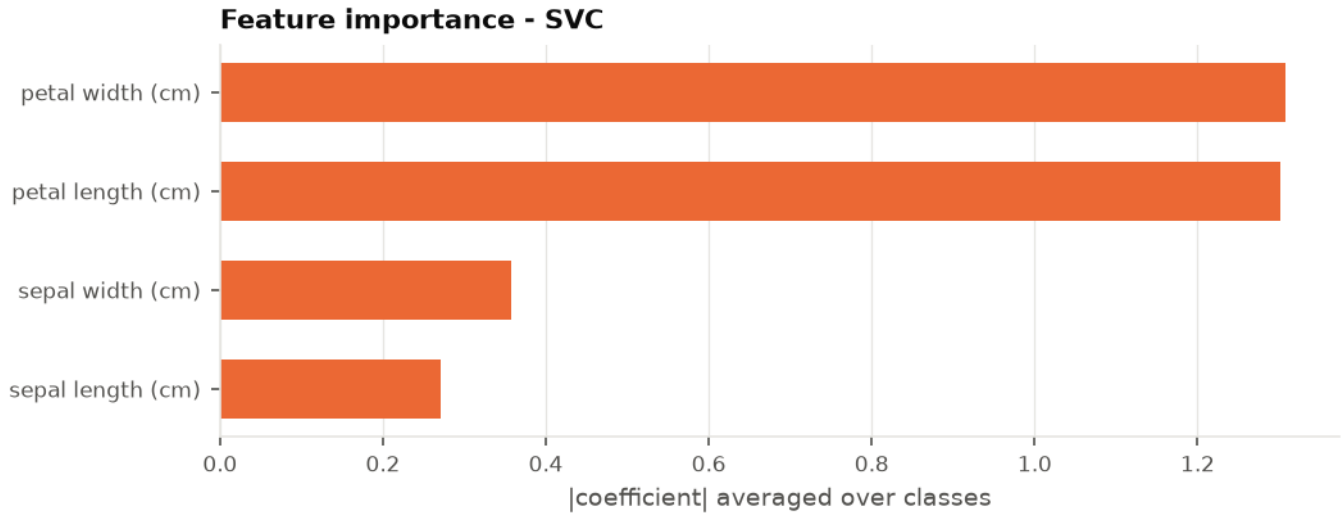


What it shows. The importance the fitted model itself assigns to each input: feature_importances_ for tree ensembles, the absolute coefficient for linear models. Shows what the model relies on, not causal effects; correlated features share importance. For a model-agnostic, held-out measure use FeatureResearch.permutation_importance().

petal length (cm)	3.3879
petal width (cm)	3.2230
sepal width (cm)	1.2619
sepal length (cm)	1.1465

Feature importance - SVC

id: feature_importance:SVC · model(s): SVC (rank #2)



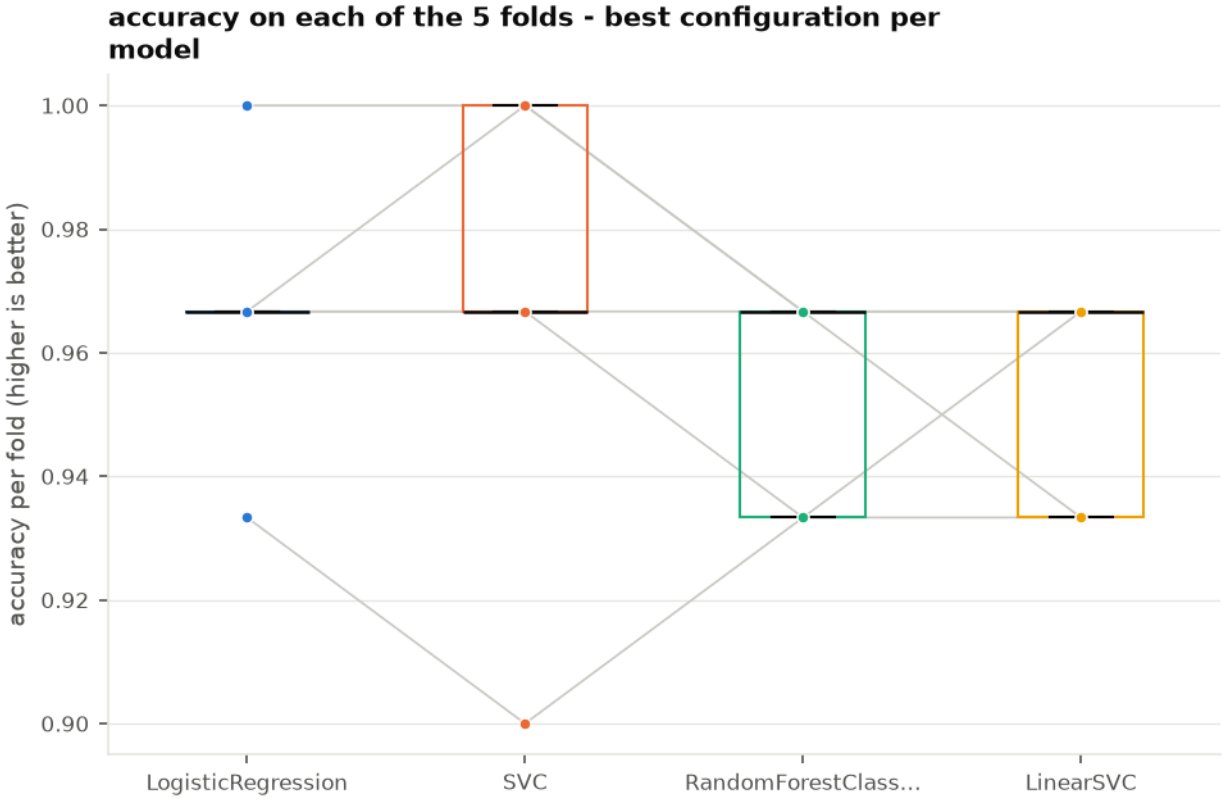
What it shows. The importance the fitted model itself assigns to each input: feature_importances_ for tree ensembles, the absolute coefficient for linear models. Shows what the model relies on, not causal effects; correlated features share importance. For a model-agnostic, held-out measure use FeatureResearch.permutation_importance().

petal width (cm)	1.3081
petal length (cm)	1.3009
sepal width (cm)	0.3574
sepal length (cm)	0.2712

7. Cross-Validation Analysis

Metric distribution across cross-validation folds

id: cv_folds · model(s): LogisticRegression (rank #1), SVC (rank #2), RandomForestClassifier (rank #6), LinearSVC (rank #9)



What it shows. The ranking metric computed separately on each validation fold for each model's best configuration. If the gap between two models is small compared with the fold-to-fold spread, the ranking is not reliable. Folds share training rows, so the spread understates the true variance and no significance test is implied.

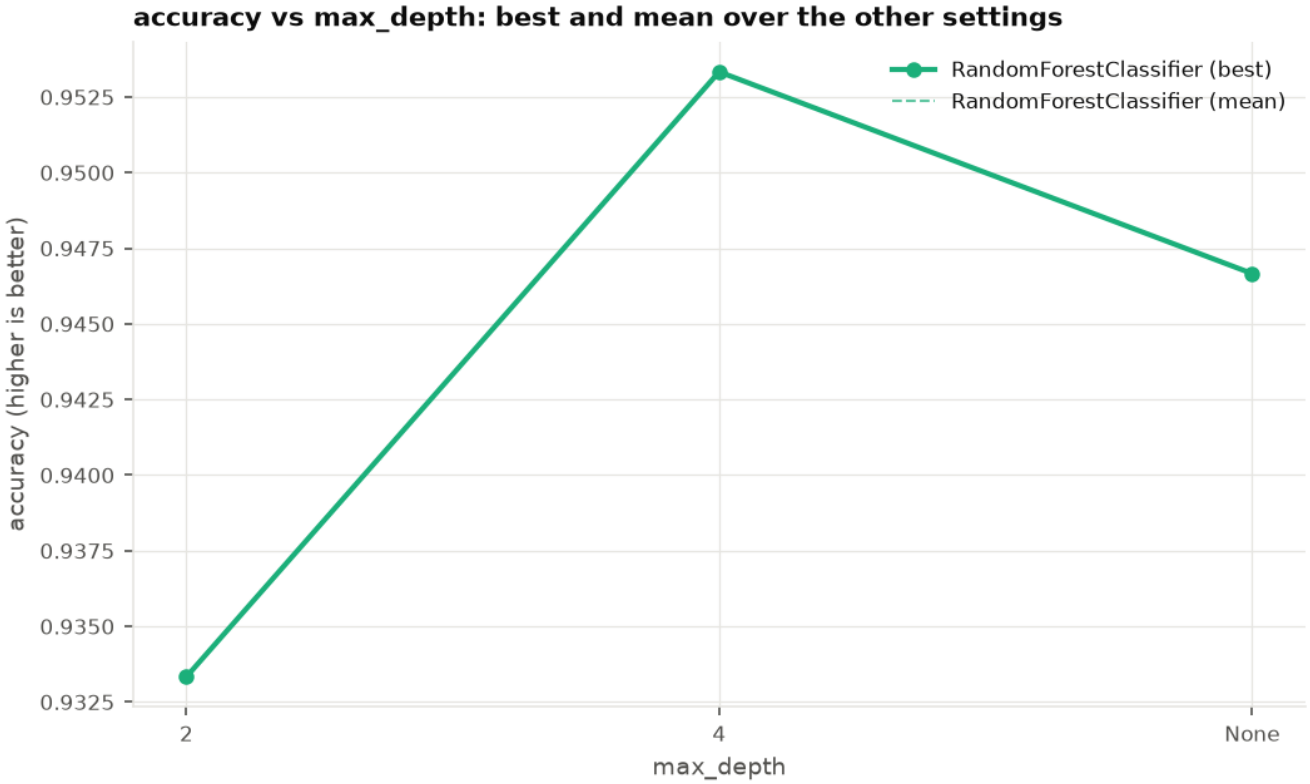
Grey lines join the same fold across models (every model is scored on identical folds).

LogisticRegression accuracy mean ± sd	0.9667 ± 0.0236
SVC accuracy mean ± sd	0.9667 ± 0.0408
RandomForestClassifier accuracy mean ± sd	0.9533 ± 0.0183
LinearSVC accuracy mean ± sd	0.9533 ± 0.0183

8. Hyperparameter Optimization

Hyper-parameter sensitivity - max_depth

id: hp_sensitivity:max_depth · model(s): RandomForestClassifier (rank #6)

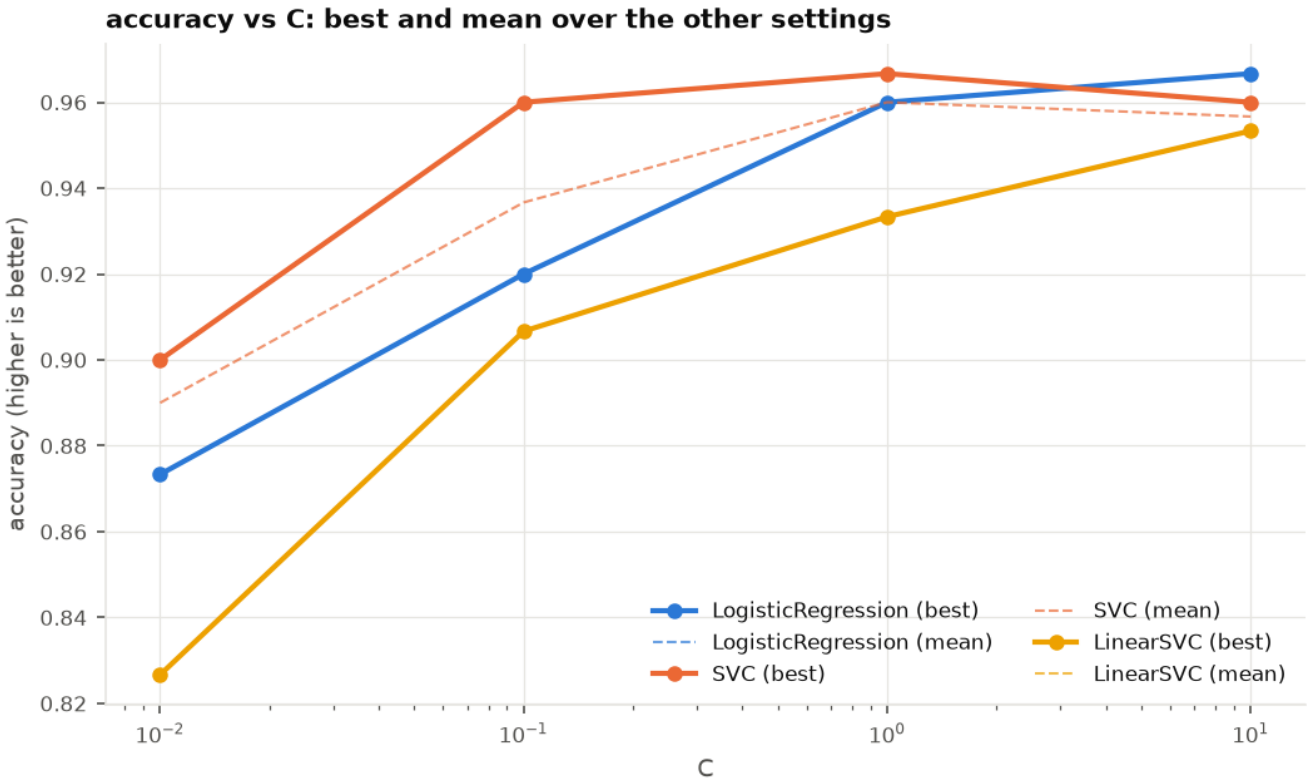


What it shows. A validation curve built from the grid already trained: for each value of one hyper-parameter, the best (solid) and mean (dashed) ranking metric over every other setting, per model. A flat curve means the parameter barely matters; a best value at the edge of the grid suggests extending the grid. Marginal view - interactions are in the heatmap.

RandomForestClassifier best max_depth	4 (accuracy 0.9533)

Hyper-parameter sensitivity - C

id: hp_sensitivity:C · model(s): LogisticRegression (rank #1), SVC (rank #2), LinearSVC (rank #9)

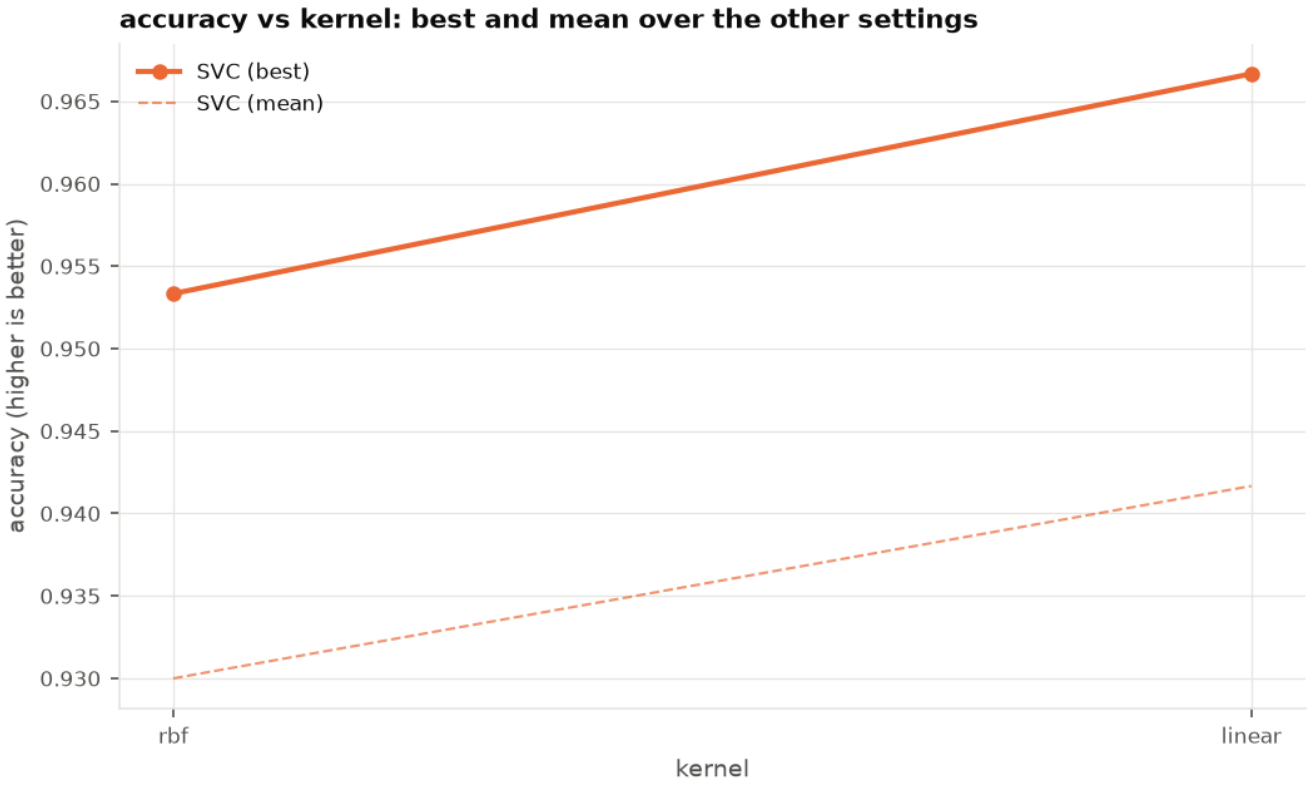


What it shows. A validation curve built from the grid already trained: for each value of one hyper-parameter, the best (solid) and mean (dashed) ranking metric over every other setting, per model. A flat curve means the parameter barely matters; a best value at the edge of the grid suggests extending the grid. Marginal view - interactions are in the heatmap.

LogisticRegression best C	10 (accuracy 0.9667)
SVC best C	1 (accuracy 0.9667)
LinearSVC best C	10 (accuracy 0.9533)

Hyper-parameter sensitivity - kernel

id: hp_sensitivity:kernel · model(s): SVC (rank #2)

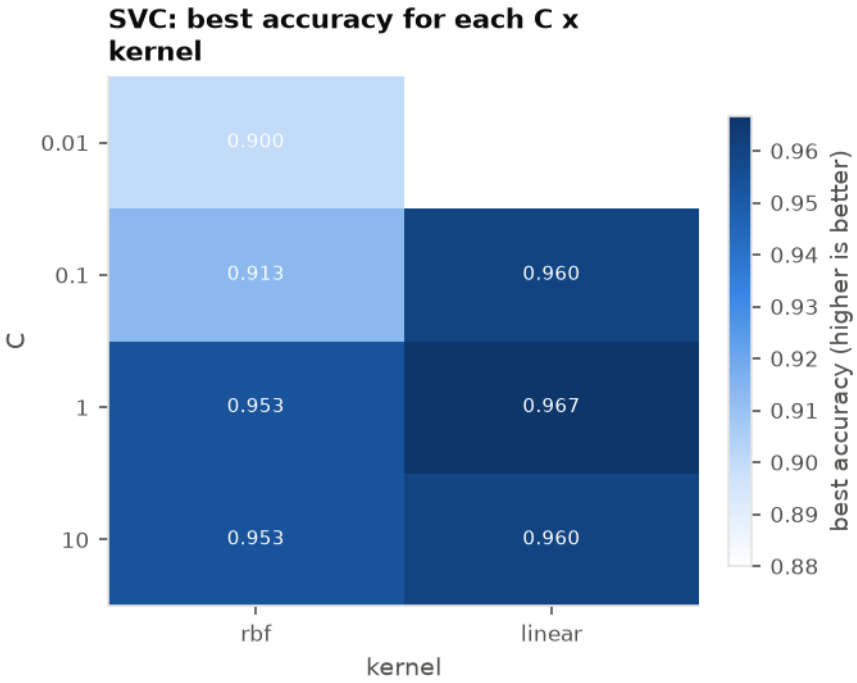


What it shows. A validation curve built from the grid already trained: for each value of one hyper-parameter, the best (solid) and mean (dashed) ranking metric over every other setting, per model. A flat curve means the parameter barely matters; a best value at the edge of the grid suggests extending the grid. Marginal view - interactions are in the heatmap.

SVC best kernel	linear (accuracy 0.9667)

Two-parameter interaction - SVC

id: hp_heatmap:SVC · model(s): SVC (rank #2)



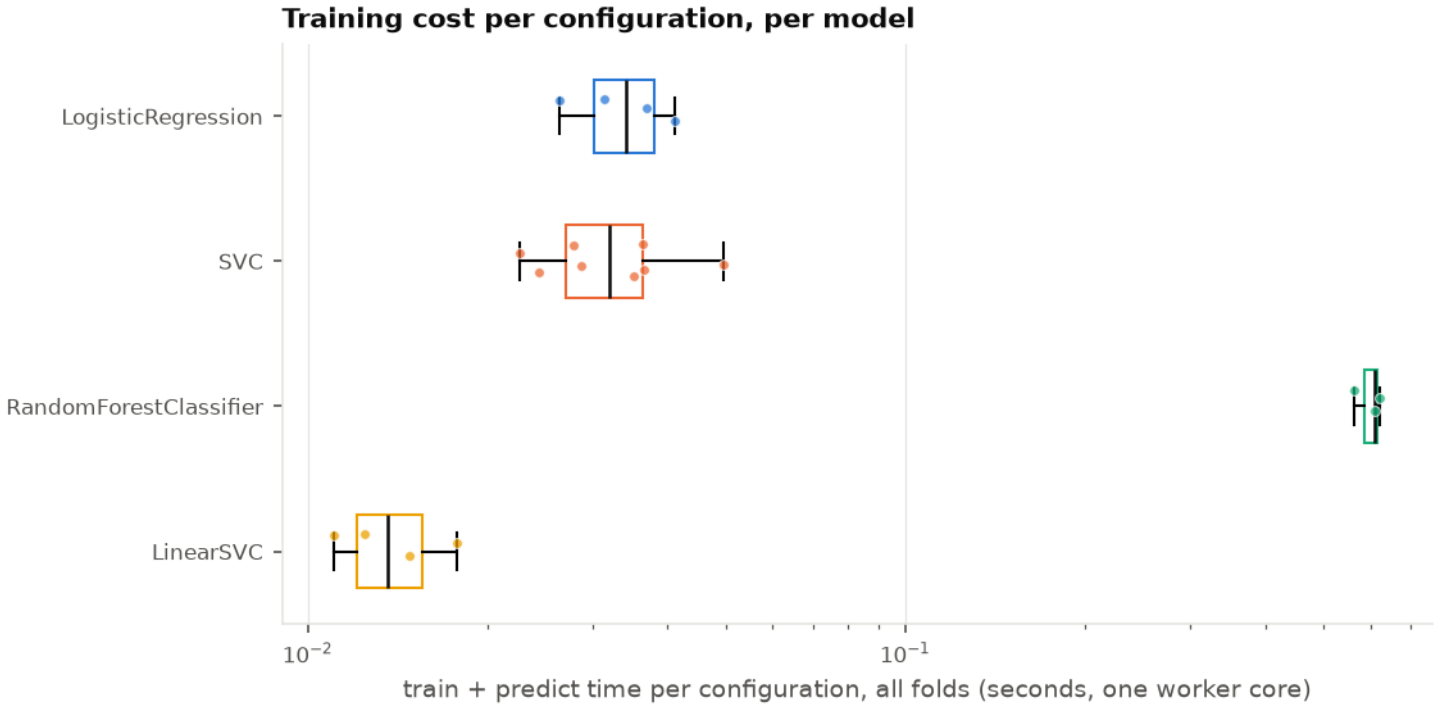
What it shows. Best ranking metric for every combination of the model's two most-varied hyper-parameters (the best over any other settings in each cell). Diagonal ridges reveal interactions that the one-parameter curves cannot show. Darker = larger value; check the colour-bar direction note.

best cell	C=1, kernel=linear (accuracy 0.9667)

9. Runtime and Resource Analysis

Runtime comparison

id: runtime · model(s): LogisticRegression (rank #1), SVC (rank #2), RandomForestClassifier (rank #6), LinearSVC (rank #9)

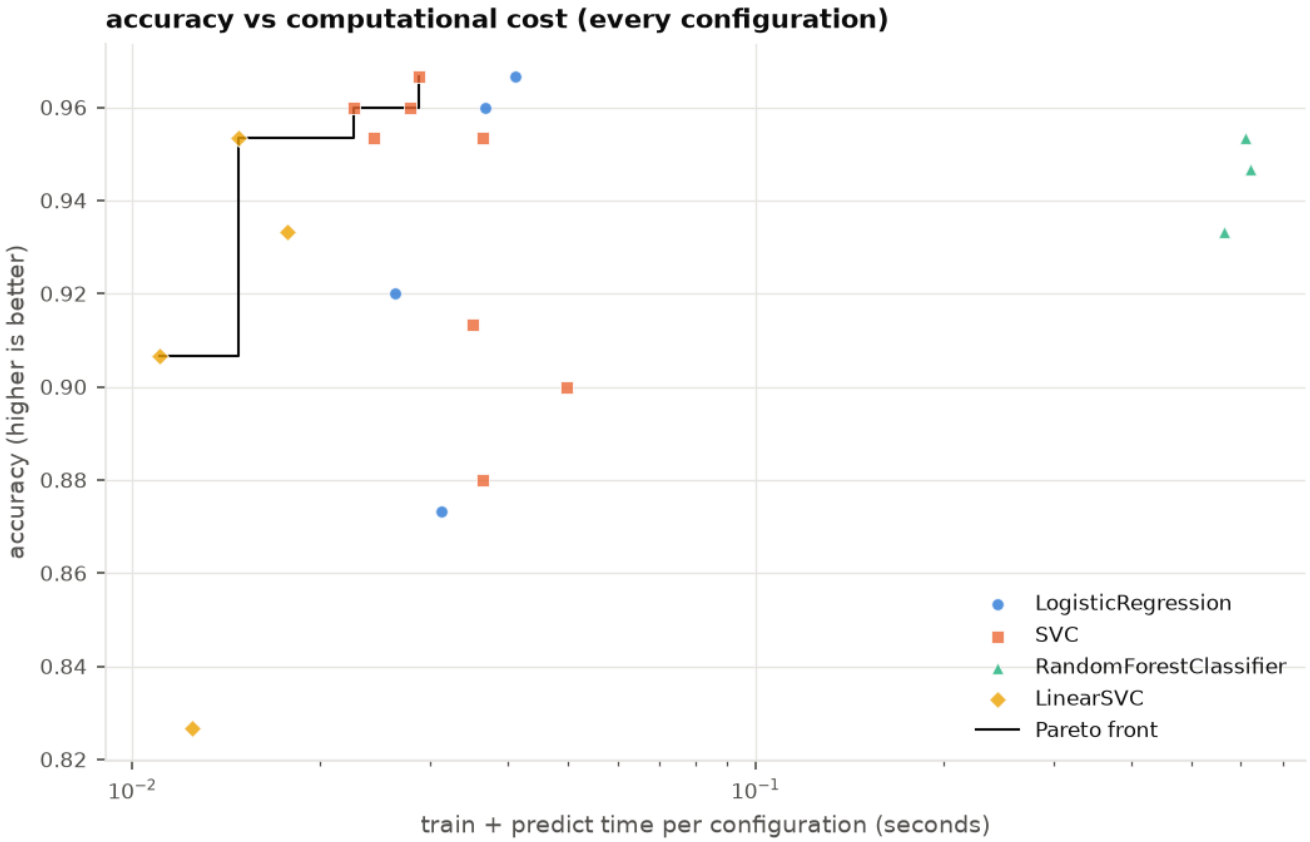


What it shows. Wall-clock time of each configuration (fit and predict over every fold), measured inside the worker process on one core. Includes preprocessing inside the pipeline; excludes data transfer. Absolute values depend on the machine and its load; compare models within one run.

LogisticRegression median / total	0.034 s / 0.14 s
SVC median / total	0.032 s / 0.26 s
RandomForestClassifier median / total	0.609 s / 1.79 s
LinearSVC median / total	0.014 s / 0.06 s

Performance vs computational cost

id: cost_vs_score · model(s): LogisticRegression (rank #1), SVC (rank #2), RandomForestClassifier (rank #6), LinearSVC (rank #9)



What it shows. Every configuration's score against its training time. The Pareto front joins the configurations that no other one beats on both score and time: a configuration near the top of the front but much further left is almost as good for a fraction of the cost.

configurations on the Pareto front	4
fastest configuration	0.011 s (accuracy 0.9067)
best configuration	0.029 s (accuracy 0.9667)

10. Visualization Index

Every visualization attempted for this report: 23 generated, 3 skipped, 0 failed.

#	visualization	section	status	time (s)	detail
1	Model ranking	Overall Results and Model Comparison	SUCCESS	0.12	included in the report
2	Score distribution across configurations	Overall Results and Model Comparison	SUCCESS	0.13	included in the report
3	Metric distribution across cross-validation folds	Cross-Validation Analysis	SUCCESS	0.14	included in the report
4	Hyper-parameter sensitivity - max_depth	Hyperparameter Optimization	SUCCESS	0.14	included in the report
5	Hyper-parameter sensitivity - C	Hyperparameter Optimization	SUCCESS	0.31	included in the report
6	Hyper-parameter sensitivity - kernel	Hyperparameter Optimization	SUCCESS	0.11	included in the report
7	Two-parameter interaction - SVC	Hyperparameter Optimization	SUCCESS	0.11	included in the report
8	Feature selection: score vs number of features	Feature Analysis	SKIPPED	0.00	every configuration used the same number of features
9	Runtime comparison	Runtime and Resource Analysis	SUCCESS	0.12	included in the report
10	Performance vs computational cost	Runtime and Resource Analysis	SUCCESS	0.17	included in the report
11	Memory consumption comparison	Runtime and Resource Analysis	SKIPPED	0.00	memory consumption is not measured per configuration by the pipeline; no values are estimated
12	Classification metrics by model	Overall Results and Model Comparison	SUCCESS	0.24	included in the report
13	Class distribution	Classification Analysis	SUCCESS	0.07	included in the report
14	Confusion matrix - LogisticRegression	Classification Analysis	SUCCESS	0.13	included in the report
15	Confusion matrix - SVC	Classification Analysis	SUCCESS	0.12	included in the report
16	Confusion matrix - RandomForestClassifier	Classification Analysis	SUCCESS	0.12	included in the report
17	Confusion matrix - LinearSVC	Classification Analysis	SUCCESS	0.12	included in the report
18	Per-class precision, recall and F1	Classification Analysis	SUCCESS	0.07	included in the report
19	Per-class F1 across models	Classification Analysis	SUCCESS	0.13	included in the report
20	ROC curves	Classification Analysis	SUCCESS	0.12	included in the report
21	ROC per class	Classification Analysis	SUCCESS	0.13	included in the report
22	Precision-recall curves	Classification Analysis	SUCCESS	0.11	included in the report
23	Precision-recall per class	Classification Analysis	SUCCESS	0.11	included in the report
24	Calibration	Classification Analysis	SKIPPED	0.00	calibration diagram is drawn for binary targets only
25	Feature importance - LogisticRegression	Feature Analysis	SUCCESS	0.08	included in the report
26	Feature importance - SVC	Feature Analysis	SUCCESS	0.08	included in the report

11. Errors, Warnings and Missing Visualizations

Skipped visualizations (3)

Not applicable to the data or results available - nothing was estimated to fill the gap.

Feature selection: score vs number of features: every configuration used the same number of features

Memory consumption comparison: memory consumption is not measured per configuration by the pipeline; no values are estimated

Calibration: calibration diagram is drawn for binary targets only